Uego, Francis Loise M. BSIT 2-1N COMP009 OOP Assignment #1

1. What is Java Programming language?
- Java is a commonly used programming language, is object oriented, mutiplatform, and reliable. It's used for game development, cloud computing, AI or artificial intelligence, and more. It's platform independent, has automatic garbage collection, versatile, and etc.

2. Explain the concept of platform independence in Java. How does Java achieve platform independence?
- The Java compiled code (byte code) can run on all platforms. A program is written in human-readable form, similar to C or most programming languages. A compiler turns that code into a format understandable by machines, and run by the JVM (Java virtual machine). The code becomes a .class file after compilation, then the JVM runs the code.

3. Discuss the main components of a Java program and their roles.
- class definition - a Java program starts with a class declaration, with the class name having the same name as the program file.
- Main method - the class with the main method is the program, and is where the program would start and stop.
- Variables - names that holds a type (int, char, string, bool, float, etc.)
- Java statements - statements that are used in Java.

4. What are the differences in JDK, JRE, and JVM in Java?
- JDK (Java Development Kit) provides the tools and libraries for JRE and JVM.
- JRE (Java Runtime Environment) offers the libraries and JVM to run Java programs.
- JVM (Java Virtual Machine) executes the compiled Java code (.class file).

5. Explain the difference between interpreted and compiled languages. Is Java an interpreted or compiled language?

- A compiler translates the entire code into machine-readable language before execution. An interpreter translates and executes code line by line.
- Java is both interpreted and compiled. It first compiles it into bytecode (.class) using javac, then interpreted by an intepreter into machine code. This makes Java platform independent.